SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I H. Divyaa Dharshini (192426014) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: H. Dhaadhini

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	1	1	2	1	1
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1	1	0	1	1

25/30

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
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Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

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Criteria	**Weightage (%)**	Q1 8 marks	Q2 8 marks	Q3 8 marks	Q4 8 marks	Q5 8 marks
Understanding of Scenario	20%					
Identification of Key Issues	20%					
Application of Concepts	25%					
Decision Making	20%					
Clarity of Response	15%					

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

Answer (8 Marks)

Given: Block 192.168.100.0/24 (256 addresses). School of Computing = /25, School of Electronics = /26, School of Mechanical Engineering = /27.

Method: Block size = 2^(32 − prefix); usable hosts = block size − 2; network address = first address of the block; broadcast address = last address of the block. Subnets are allocated largest first, one after another, so they do not overlap.

Step 1: Masks and block sizes

Department	Prefix	Subnet mask	Block size	Usable hosts
School of Computing	/25	255.255.255.128	128	2^7 − 2 = 126
School of Electronics	/26	255.255.255.192	64	2^6 − 2 = 62
School of Mech. Engg.	/27	255.255.255.224	32	2^5 − 2 = 30

Step 2: Allocation

Computing: starts at .0, covers 128 addresses (.0 to .127)

Electronics: next free = .128, covers 64 addresses (.128 to .191)

Mechanical: next free = .192, covers 32 addresses (.192 to .223)

Step 3: Result

Department	Network address	Valid host range	Broadcast	Usable hosts
Computing	192.168.100.0/25	192.168.100.1 – 192.168.100.126	192.168.100.127	126
Electronics	192.168.100.128/26	192.168.100.129 – 192.168.100.190	192.168.100.191	62
Mechanical	192.168.100.192/27	192.168.100.193 – 192.168.100.222	192.168.100.223	30

Step 4: Unused range. Addresses used = 128 + 64 + 32 = 224 (192.168.100.0 to 192.168.100.223). The unused range is 192.168.100.224 – 192.168.100.255, which is 256 − 224 = 32 addresses (equal to the block 192.168.100.224/27).

Interpretation: The three subnets are contiguous and non-overlapping and cover 87.5% of the /24 block. The free /27 can be kept for a future department. Total usable hosts = 126 + 62 + 30 = 218.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.

3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

Answer (8 Marks)

Given: Private block 10.10.0.0/16 (65,536 addresses). Software Development = /24, Quality Assurance = /25, Human Resources = /26, Executive Management = /27.

Method: Allocate the largest block first, each new subnet starting where the previous one ended. Usable hosts = 2^(32 − prefix) − 2.

Step 1: Masks and block sizes

Department	Prefix	Subnet mask	Block size	Usable hosts
Software Development	/24	255.255.255.0	256	2^8 − 2 = 254
Quality Assurance	/25	255.255.255.128	128	2^7 − 2 = 126
Human Resources	/26	255.255.255.192	64	2^6 − 2 = 62
Executive Management	/27	255.255.255.224	32	2^5 − 2 = 30

Step 2: Allocation

Software Development: 10.10.0.0 to 10.10.0.255 (256 addresses)

Quality Assurance: next free = 10.10.1.0, covers 128 (10.10.1.0 to 10.10.1.127)

Human Resources: next free = 10.10.1.128, covers 64 (to 10.10.1.191)

Executive Management: next free = 10.10.1.192, covers 32 (to 10.10.1.223)

Step 3: Result

Department	Network address	Valid host range	Broadcast	Usable hosts
Software Dev.	10.10.0.0/24	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	10.10.1.0/25	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	10.10.1.128/26	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Mgmt.	10.10.1.192/27	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Step 4: Unused range. Addresses used = 256 + 128 + 64 + 32 = 480 (10.10.0.0 to 10.10.1.223). The unused range is 10.10.1.224 – 10.10.255.255, which is 65,536 − 480 = 65,056 addresses.

Interpretation: All four subnets are contiguous, non-overlapping and sit within 10.10.0.0/23. Most of the /16 block is still free for future growth. Total usable hosts = 254 + 126 + 62 + 30 = 472.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

Answer (8 Marks)

Given: IPv6 network 2001:0db8:abcd:0000:0000:0000:0000:0000/64 (128 bits, eight 16-bit groups).

Task 1: Compressed notation

Rule 1: leading zeros in each group are dropped. Rule 2: one run of consecutive all-zero groups is replaced by '::' (only once).

Full: 2001:0db8:abcd:0000:0000:0000:0000:0000

Rule 1: 2001:db8:abcd:0:0:0:0:0

Rule 2: 2001:db8:abcd::

Compressed form: 2001:db8:abcd::/64

Task 2: Expansion

The compressed form has 3 groups written (2001, db8, abcd), so 8 − 3 = 5 zero groups are restored by '::', and each group is padded to 4 hex digits.

2001:db8:abcd:: → 2001:0db8:abcd:0000:0000:0000:0000:0000

Expanded form: 2001:0db8:abcd:0000:0000:0000:0000:0000 (same as the given address).

Task 3: Network prefix

With /64, the first 64 bits (the first four groups, 2001:0db8:abcd:0000) form the network prefix. The last 64 bits are the interface identifier (host part).

Network prefix: 2001:db8:abcd:0::/64, written 2001:db8:abcd::/64.

Tasks 4 and 5: Number of host addresses

Tasks 4 and 5 ask for the same quantity, so they are answered together. Host bits = 128 − 64 = 64, so the total number of addresses = 2^64 = 18,446,744,073,709,551,616 (about 1.84 × 10^19).

Interpretation: A /64 is the standard IPv6 subnet size and is far more than enough for thousands of medical devices and IoT sensors in the hospital. IPv6 has no broadcast address, so no address is lost to broadcast; only the all-zeros address 2001:db8:abcd:: (subnet-router anycast) is conventionally reserved, giving 2^64 − 1 assignable host addresses if it is excluded.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

Answer (8 Marks)

Given: Routing table entries 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24. Destination of the packet = 192.168.2.45.

Method: The router compares the destination address with each entry using the subnet mask and forwards the packet using the longest prefix match. All entries are /24, so the mask is 255.255.255.0 and the first three octets must match.

Task 1: Destination subnet

192.168.2.45 AND 255.255.255.0 = 192.168.2.0

Subnet 192.168.2.0/24: network 192.168.2.0, hosts 192.168.2.1 – 192.168.2.254, broadcast 192.168.2.255

The destination 192.168.2.45 lies in the range of 192.168.2.0/24 (host number 45 is within 1 to 254).

Task 2: Matching routing table entry

Routing table entry	Network ID after AND with mask	Match?
192.168.1.0/24	192.168.2.0 ≠ 192.168.1.0	No
192.168.2.0/24	192.168.2.0 = 192.168.2.0	Yes (prefix length 24)
192.168.3.0/24	192.168.2.0 ≠ 192.168.3.0	No

The matching entry is 192.168.2.0/24. It is the only match, so it is also the longest prefix match.

Task 3: Next hop / outgoing interface

The packet is forwarded out of the router interface connected to the 192.168.2.0/24 customer network. The question does not name the interface, so it is assumed to be the router port facing that network (for example Gi0/1, with the router address 192.168.2.1 on it). If the network is directly connected, the router delivers the packet straight to host 192.168.2.45 using ARP; if the customer network is reached through another router, the next-hop address listed for this entry is used.

Task 4: Default route

If the destination matches none of the entries (for example 8.8.8.8), the router uses the default route 0.0.0.0/0, which matches every address with prefix length 0 and is chosen only when no longer prefix matches. It points to the ISP upstream router (next hop or exit interface), for example:

ip route 0.0.0.0 0.0.0.0 <upstream next-hop IP or exit interface>

Interpretation: The packet 192.168.2.45 is forwarded using the 192.168.2.0/24 entry. Packets to unknown destinations are sent upstream by the default route 0.0.0.0/0, and if no default route exists, the router drops them and returns an ICMP destination-unreachable message.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

Answer (8 Marks)

Given: Administration LAN 192.168.10.0/26 (gateway 192.168.10.1) and Finance LAN 192.168.10.64/26 (gateway 192.168.10.65), connected through a Layer-3 router.

Method: /26 = 255.255.255.192, host bits = 6, block size = 2^6 = 64, usable hosts = 64 − 2 = 62. Network address = first address, broadcast address = last address of each block.

Tasks 1 and 2: Network, broadcast and host range

Parameter	Administration LAN	Finance LAN
Network address	192.168.10.0/26	192.168.10.64/26
Subnet mask	255.255.255.192	255.255.255.192
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Usable hosts	62	62
Router gateway	192.168.10.1	192.168.10.65

Task 3: Suggested computer addresses

LAN	Computer	IP address	Subnet mask	Default gateway
Administration	PC1	192.168.10.2	255.255.255.192	192.168.10.1
Administration	PC2	192.168.10.3	255.255.255.192	192.168.10.1
Administration	PC3	192.168.10.4	255.255.255.192	192.168.10.1
Finance	PC1	192.168.10.66	255.255.255.192	192.168.10.65
Finance	PC2	192.168.10.67	255.255.255.192	192.168.10.65
Finance	PC3	192.168.10.68	255.255.255.192	192.168.10.65

Each address is a valid host address in its own subnet and differs from the network, broadcast and gateway addresses.

Task 4: Default gateway

Administration LAN hosts: 192.168.10.1 (the router interface in 192.168.10.0/26).

Finance LAN hosts: 192.168.10.65 (the router interface in 192.168.10.64/26).

Interpretation: The two LANs are different subnets, so a host in one cannot reach the other directly. A host sends traffic for another subnet to its default gateway, and the router forwards it to the other LAN. The gateway must always be in the same subnet as the host, which is why each LAN uses its own gateway address.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
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